

ILIRNEY, Russian Federation

WMO: 252480

Lat: 67.256N

Lon: 167.962E

Elev: 354

StdP: 97.15

Time Zone: 12.00 (E12)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-49.7	-47.9	-52.1	0.0	-49.1	-50.3	0.0	-47.8	9.2	-9.2	8.3	-14.9	0.0	130	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	25.1	15.0	22.8	14.1	20.6	13.2	16.3	22.6	15.2	21.0	13.9	19.0	2.3	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.1	10.5	18.0	12.6	9.5	16.8	11.2	8.6	16.2	47.1	22.8	43.7	21.1	40.3	19.0	22.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.9	6.5	5.7	-52.1	28.3	2.2	2.3	-53.7	29.9	-55.0	31.2	-56.3	32.5	-57.9	34.2		
(5)			-52.0	18.6	2.1	1.9	-53.6	20.0	-54.8	21.1	-56.0	22.2	-57.6	23.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-11.1	-33.8	-32.1	-25.7	-15.1	-1.0	10.9	12.8	9.7	2.9	-10.2	-23.1	-30.2		
	DBStd	18.52	9.97	9.25	8.15	6.92	6.14	4.25	4.05	4.00	4.30	7.59	9.15	10.08		
	HDD10.0	7931	1357	1179	1106	752	341	42	19	55	216	627	993	1245		
	HDD18.3	10754	1615	1412	1364	1002	598	224	176	268	463	885	1243	1503		
	CDD10.0	222	0	0	0	0	1	68	105	46	3	0	0	0		
	CDD18.3	5	0	0	0	0	0	1	4	1	0	0	0	0		
	CDH23.3	117	0	0	0	0	0	28	81	8	0	0	0	0		
	CDH26.7	14	0	0	0	0	0	2	11	2	0	0	0	0		
(14) Wind	WSAvg	1.6	1.1	1.0	1.5	1.6	2.2	2.3	2.1	2.0	2.0	1.4	1.4	1.1		
(15) Precipitation	PrecAvg	191	9	10	10	5	8	18	26	28	17	20	17	12		
	PrecMax	296	35	42	28	20	20	51	67	91	48	82	43	34		
	PrecMin	93	0	0	0	0	1	2	4	5	2	3	0	0		
	PrecStd	55	8	10	8	5	5	11	16	17	11	18	10	9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.5	-0.7	-0.3	2.7	18.3	27.0	29.1	24.2	19.6	7.2	-0.5	-2.2		
		MCWB	-2.6	-1.8	-1.7	-0.5	9.4	14.4	16.8	15.1	11.3	5.8	-1.4	-3.0		
	2%	DB	-5.9	-7.0	-4.5	0.4	13.7	24.1	26.5	21.7	15.1	3.3	-2.7	-6.3		
		MCWB	-6.8	-7.8	-5.8	-2.0	7.2	13.2	16.2	13.9	9.1	2.3	-3.6	-7.2		
	5%	DB	-12.1	-12.9	-8.8	-1.3	10.7	22.1	24.0	19.8	12.5	1.4	-5.3	-9.8		
		MCWB	-12.8	-13.5	-9.6	-3.3	5.7	13.1	15.1	12.8	8.0	0.5	-6.0	-10.4		
	10%	DB	-19.5	-17.8	-13.3	-3.3	8.0	19.9	21.2	17.5	10.1	-0.3	-8.3	-15.3		
		MCWB	-19.9	-18.1	-14.3	-5.0	4.1	12.0	14.0	12.0	6.7	-1.2	-8.8	-15.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.5	-2.3	-1.6	0.2	9.6	16.3	19.0	17.3	12.0	5.8	-1.2	-3.1		
		MCDB	-1.6	-1.4	-0.5	2.6	16.9	24.0	25.4	21.6	17.7	7.2	-0.4	-2.1		
	2%	WB	-6.6	-7.8	-5.7	-1.9	7.5	14.5	17.0	15.2	10.3	2.5	-3.6	-7.2		
		MCDB	-5.9	-7.0	-4.6	0.2	13.2	21.5	23.3	19.9	13.9	3.4	-2.7	-6.3		
	5%	WB	-12.7	-13.6	-9.8	-3.3	5.8	13.3	15.9	13.7	8.5	0.6	-6.0	-10.5		
		MCDB	-12.0	-13.0	-8.9	-1.4	10.5	20.8	22.4	18.1	11.8	1.4	-5.3	-9.7		
	10%	WB	-19.8	-18.1	-14.3	-5.0	4.2	12.4	14.9	12.5	7.0	-1.2	-8.8	-15.8		
		MCDB	-19.5	-17.8	-13.4	-3.4	7.8	19.4	20.6	16.9	9.9	-0.3	-8.3	-15.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.1	9.8	13.1	15.0	9.8	10.8	10.1	10.9	8.9	7.9	8.7	8.2		
		MCDBR	12.6	11.4	11.6	13.7	12.6	15.9	15.8	15.4	14.6	5.2	8.7	12.6		
		MCWBR	12.2	11.1	10.5	11.8	7.4	7.6	7.4	8.2	9.1	4.1	8.3	12.2		
		MCDBR	12.5	11.1	11.3	13.2	12.3	14.2	12.6	12.5	12.3	4.8	9.3	12.3		
	5% WB	MCWBR	12.1	10.9	10.5	11.4	7.4	7.5	7.2	7.6	8.5	4.0	9.0	11.9		
(40) Clear-Sky Solar Irradiance	taub		0.155	0.219	0.284	0.333	0.370	0.357	0.404	0.384	0.303	0.245	0.172	N/A		
	taud		1.931	2.124	2.087	1.991	2.055	2.382	2.257	2.273	2.451	2.162	1.920	N/A		
	Ebn at Noon		225	636	750	801	812	847	781	753	756	569	161	0		
	Edh at Noon		11	47	88	129	136	102	112	99	64	45	10	0		
(44) All-Sky Solar Radiation	RadAvg		0.05	0.58	2.37	4.71	5.80	5.48	4.81	3.59	1.98	0.75	0.12	0.00		
	RadStd		0.01	0.07	0.16	0.17	0.43	0.36	0.51	0.37	0.19	0.09	0.01	0.00		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page